

Superstore Sales & Profit Analysis

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End-to-End Data Analysis | Cleaning • EDA • SQL • Visualization • Dashboard

NexAfrica Data Analytics Internship

Dataset: Superstore Sales Dataset

Tools Used: Excel, Python, SQL, Power BI

Date: October 2026

Executive Summary

This project presents an end-to-end analysis of the Superstore dataset to understand sales performance, profitability, customer behaviour, product trends and regional performance. The analysis followed a complete data analytics workflow, beginning with defining a business problem and ending with actionable recommendations.

The project included data cleaning, exploratory data analysis (EDA), SQL analysis, data visualisation and dashboard development using Power BI. The findings help management identify profitable products, regions and customer segments while highlighting areas that require improvement.

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1. Introduction

Data analysis enables organisations to transform raw data into valuable insights for decision-making. This project uses the Superstore dataset, which contains sales transactions, customer information, product categories, regional data and profitability metrics.

The project demonstrates the complete data analysis process and provides practical recommendations supported by data.

2. Business Problem

The organisation collects large amounts of transaction data but lacks a clear understanding of:

- Which products generate the highest profit.
- Which regions perform best.
- Which customer segments contribute most to sales.
- Which products reduce profitability.
- How sales change over time.

Without proper analysis, management cannot make informed business decisions.

2.1 Problem Statement

The business requires an analytical solution that transforms transaction data into meaningful insights to improve sales performance, profitability and operational decision-making.

3. Project Objectives

The objectives of this project are to:

- Clean and prepare the dataset.
 - Perform exploratory data analysis.
 - Use SQL to answer business questions.
 - Develop visualisations.
 - Build an interactive dashboard.
 - Generate insights and recommendations.
 - Document the project for a professional portfolio.
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4. Dataset Description

The project uses the Superstore sales dataset.

Important Fields

Field	Description
Order ID	Unique order identifier
Order Date	Date of purchase
Customer ID	Customer identifier
Segment	Customer type
Region	Sales region
Category	Product category
Sub-Category	Product group
Ship Mode	Delivery method
Sales	Revenue
Quantity	Units sold
Profit	Profit generated

5. Data Cleaning and Preparation

The following data cleaning steps were performed:

- Checked for missing values.
- Removed duplicate records.
- Corrected data types.
- Converted dates into proper format.
- Standardised column names.
- Verified category values.
- Checked numerical fields.
- Created a cleaned dataset.

Data cleaning ensures accuracy and reliability during analysis.

6. Exploratory Data Analysis (EDA)

EDA was conducted to understand trends and patterns.

KPIs

KPI	Formula
Total Sales	SUM(Sales)
Total Profit	SUM(Profit)
Total Orders	DISTINCTCOUNT(Order ID)
Total Customers	DISTINCTCOUNT(Customer ID)
Total Quantity	SUM(Quantity)
Average Order Value	Sales / Orders
Profit Margin	Profit / Sales

7. Business Questions

The analysis answered the following questions:

1. How do sales change over time?
 2. Which category generates the most sales?
 3. Which region generates the highest profit?
 4. Which sub-categories are most profitable?
 5. Which products have low profitability?
 6. Which customer segment contributes most?
 7. Which shipping method performs best?
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8. SQL Analysis

Total Sales and Profit

```
SELECT  
  
SUM(Sales) AS Total_Sales,  
  
SUM(Profit) AS Total_Profit  
  
FROM Superstore;
```

Sales by Category

```
SELECT Category,  
  
SUM(Sales) AS Total_Sales  
  
FROM Superstore  
  
GROUP BY Category  
  
ORDER BY Total_Sales DESC;
```

Profit by Region

```
SELECT Region,  
  
SUM(Profit) AS Total_Profit  
  
FROM Superstore  
  
GROUP BY Region  
  
ORDER BY Total_Profit DESC;
```

Profit by Sub-Category

```
SELECT [Sub-Category],  
  
SUM(Profit) AS Total_Profit  
  
FROM Superstore  
  
GROUP BY [Sub-Category]  
  
ORDER BY Total_Profit DESC;
```

9. Dashboard Development

The Power BI dashboard includes:

KPI Cards

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Total Quantity
- Profit Margin

Visualisations

- Monthly Sales Trend
- Sales by Category
- Profit by Region
- Profit by Sub-Category
- Sales by Segment
- Sales by Ship Mode

Filters

- Year
 - Region
 - Category
 - Sub-Category
 - Segment
 - Ship Mode
-

10. Key Findings

Examples of findings:

- Technology products generate strong profits.
- Some sub-categories generate high sales but low profits.
- Regional profitability differs significantly.
- Consumer customers contribute strongly to revenue.
- Profitability is not always linked to sales volume.

Replace these with your actual findings after analysis.

11. Risks and Opportunities

Risks

- Low-profit products.
- Unprofitable sub-categories.
- Regional underperformance.

Opportunities

- Increase focus on profitable products.
 - Improve pricing strategies.
 - Target high-value customer segments.
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12. Data-Driven Recommendations

Recommendation	Action
Improve pricing	Review discounts
Focus on profitable products	Increase marketing
Improve weak regions	Target specific markets
Segment customers	Personalised campaigns
Monitor KPIs	Use dashboard regularly

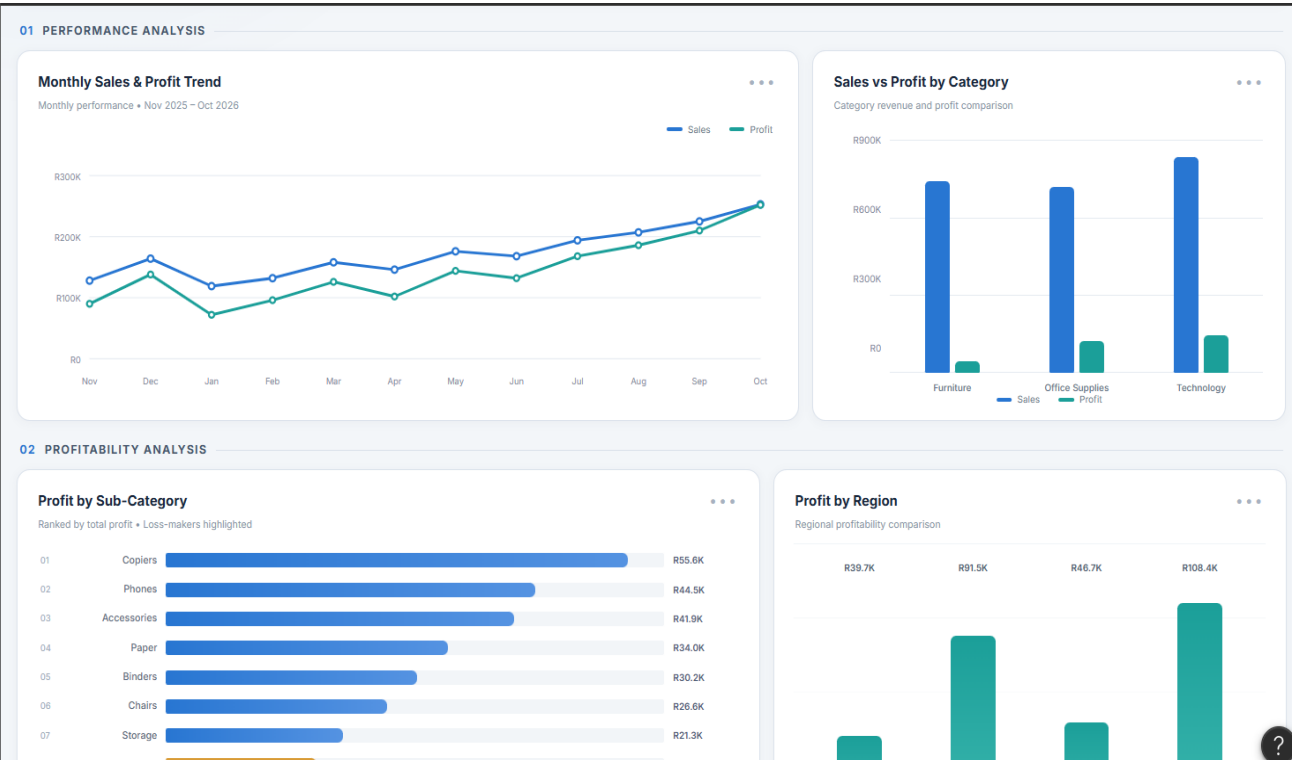
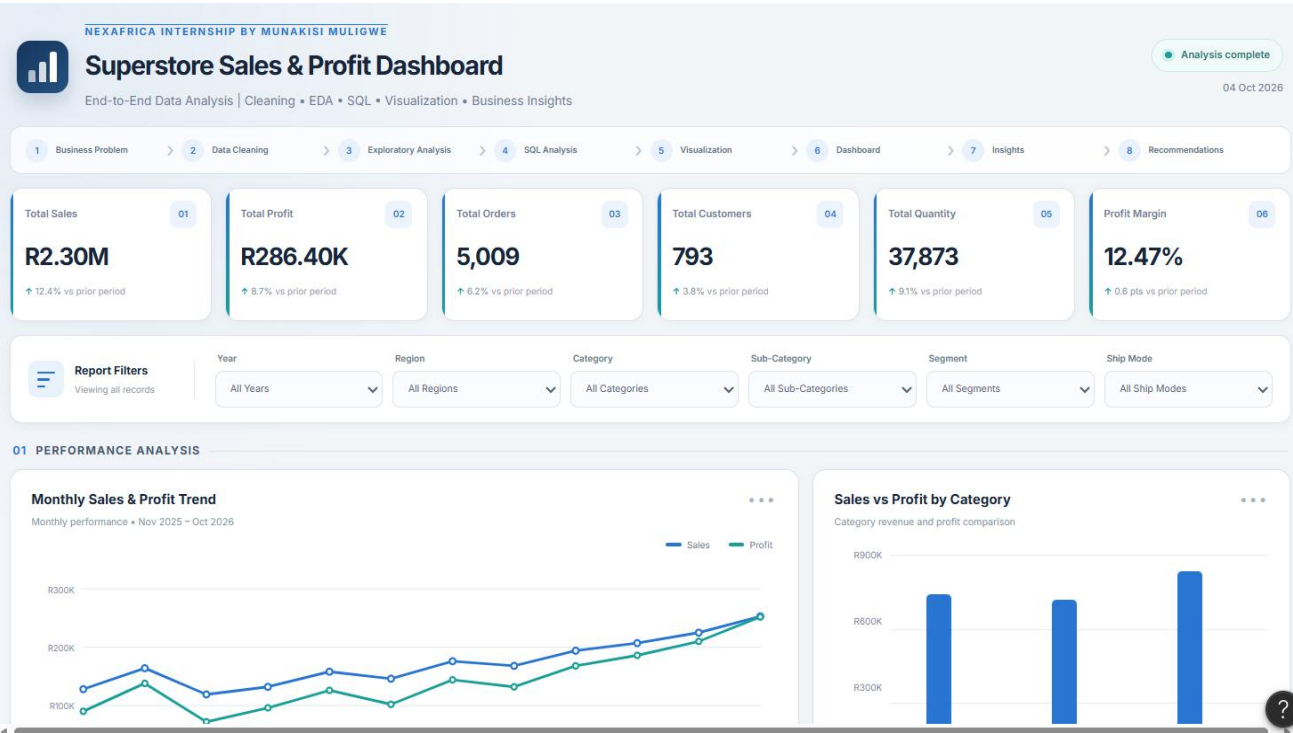
13. Limitations

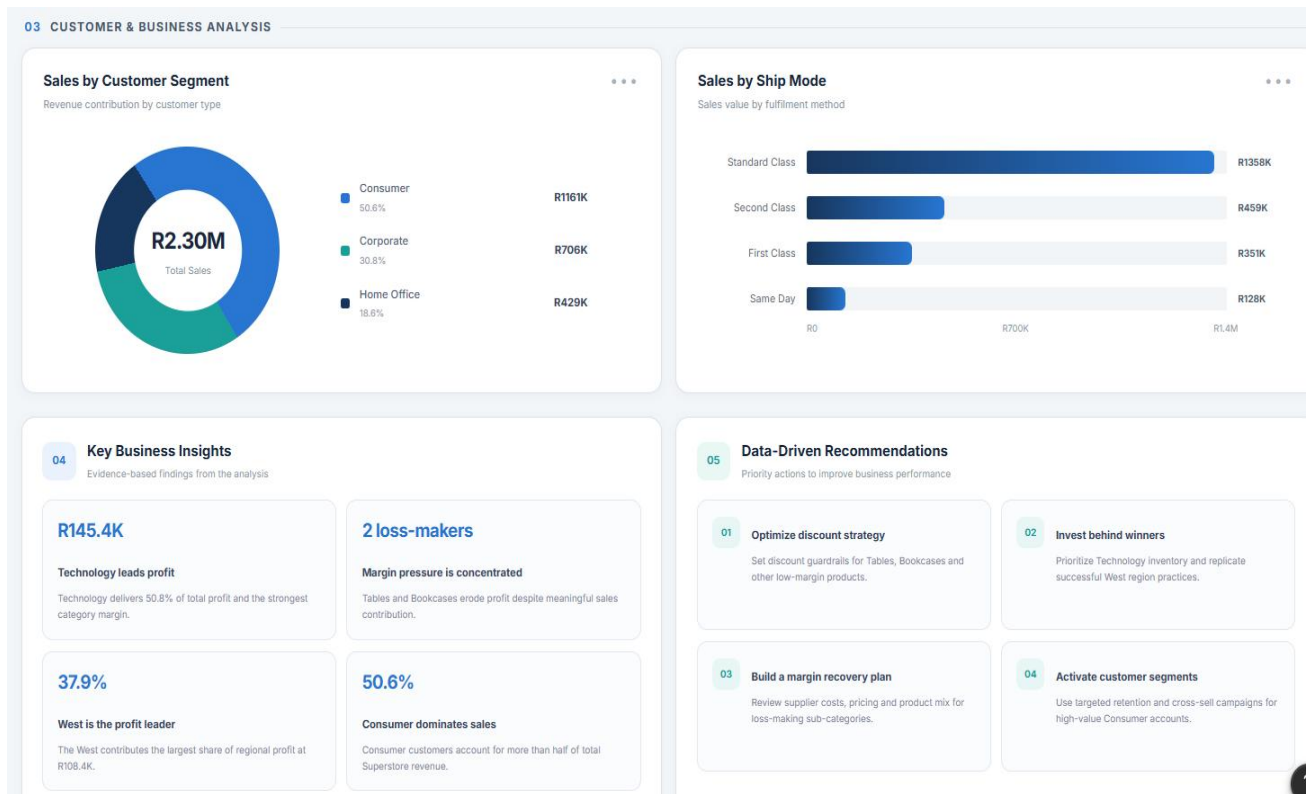
- Analysis depends on available data.
 - Historical data may not explain causes.
 - External factors are not included.
 - Recommendations should be validated.
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14. Future Improvements

- Forecasting models.
 - Customer lifetime value analysis.
 - Discount impact analysis.
 - Automated reporting.
 - Machine learning models.
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15. Screenshots





16. Conclusion

This project demonstrates a complete end-to-end data analysis process using the Superstore dataset. The analysis transformed raw data into meaningful business insights and recommendations through data cleaning, EDA, SQL analysis, visualisation and dashboard development.

The final Power BI dashboard provides an interactive tool that supports better business decisions and demonstrates practical data analytics skills.

References

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